

CS6290 Privacy-Enhancing Technologies

Group Project Instructions (AI-era Edition, Milestone-Based, Build & Test Oriented)

This guideline is designed to:

- encourage **building, testing, and empirical analysis**
- make AI usage **productive but accountable**

The project uses **3 milestone checkpoints**, each with **individual Evidence Packs only**. A **+5% bonus** is available for well-executed specification-driven development.

0) Overview

This course includes a substantial **group project**, which contributes **20%** to the final course grade.

Projects must be relevant to **privacy-enhancing technologies, blockchain, Web3, DeFi, or decentralised systems**, and must involve **building, testing, or empirical measurement**.

⚠ **Pure literature reviews / SoK-only / narrative-only survey projects are NOT accepted.**

Each group must produce:

- a **final written report**
- a **final presentation**
- a **repository link (highlighted upfront in the final written report)** containing code, scripts, specifications, or experiments
(not a ZIP upload)

1) Learning goals

By the end of the project, you should be able to:

- define a clear technical problem, assumptions, and **threat model**
- design or analyse a system with **security and privacy reasoning**
- produce **evidence-backed results** (tests, measurements, experiments)
- communicate clearly in writing and orally
- use AI tools **responsibly and critically**

2) Team formation & accountability

2.1 Team size

Recommended: **4–5 students per team**

2.2 Roles (recommended; rotate at least once)

- **Lead / PM:** scope, timeline, decisions
- **Architecture / Spec owner:** interfaces, assumptions, invariants
- **Implementation / Tooling:** smart contracts, backend, scripts
- **Security & Verification:** threat model, tests, adversarial thinking
- **Demo & Writing:** slides, report integration

2.3 Individual accountability (mandatory)

Each student must submit an **Individual Evidence Pack** for **every milestone**. There are **no team milestone submissions**.

Your individual Evidence Packs are the **primary basis** for assessing contribution and progress.

3) Responsible AI usage (mandatory)

AI tools are allowed, but **you are responsible** for correctness and integrity.

Allowed

- brainstorming ideas and alternatives
- drafting code, specs, or report sections
- summarising papers (must verify)
- generating tests or analysis scripts

Not allowed

- submitting unverified AI output as facts or results
- fabricating experiments, data, or citations
- hiding substantial AI assistance

AI Collaboration Appendix (required)

Your final report must include:

- tools used
 - what they were used for
 - **one example** where AI output was wrong/misleading and how you corrected it
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4) Project workflow (Milestone-based, individual submission)

The project uses **3 milestone checkpoints**.

Milestone loop (individual):

AI drafting → validation (self + team discussion) → evidence logging

Milestone 1 — Proposal & Plan (individual)

- problem statement + scope
- threat model v1

- validation / experiment plan
- initial specification or prototype plan

Milestone 2 — Midway Evidence (individual)

- concrete progress + early results
- threat model v2 + tests
- evidence addressing at least one non-trivial risk or uncertainty

Milestone 3 — Final Integration (individual)

- consolidated contributions
- validation results
- how your work integrates into the final report, slides, and repo

5) Project Type (Unified: Build, Test, Measure)

⚠ **There is only ONE project direction.**

Pure SoK / literature-only surveys are **not accepted**.

Your project must include **at least one** of:

- a working **MVP / prototype**
- **executable specifications** with validation
- **empirical analysis** on real or simulated data
- **security / privacy testing**

Narrative-only analysis without validation will receive **reduced credit**.

5.1 Specification-Driven Development (Core Skill)

All teams are expected to use **spec-driven development**.

Minimum expected artifacts:

- interfaces & data models
- assumptions & invariants
- acceptance criteria (Given / When / Then)
- threat → test / measurement mapping

Specs may validate:

- team-written implementations, or
- AI-assisted implementations

5.2 Example Project Topics (Strongly Recommended)

1. AI agents on blockchain — build an agent + test safety and manipulation risks
2. MVP perpetual DEX (Hyperliquid-style) — simplified matching + liquidation logic

3. Polymarket signal analysis — detect unusual odds movements empirically
 4. MEV strategy simulation and defense testing
 5. Wallet / RPC privacy leakage measurement
 6. Account abstraction (ERC-4337) security testing
 7. Cross-chain bridge attack surface experiments
 8. ZK application MVP — identity, voting, or proof-of-reserves
 9. DeFi oracle robustness under stress
 10. On-chain governance attack simulation
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6) Deliverables & submission

Final deliverables (team)

1. Final report (PDF)
2. Slides (PDF or PPTX)
3. Repository link (code / scripts / specs)

Milestones (individual only)

For **each** milestone, **every student** submits:

- **one Individual Evidence Pack**

There are **no team milestone submissions**.

7) Individual Evidence Pack (mandatory)

Format: **1–2 page PDF**

Must include:

1. What you contributed (2–5 bullets)
 2. Verifiable evidence (≥ 2 items)
 3. Validation performed (≥ 1 item)
 4. AI usage: one adopted, one rejected (with reason)
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8) Presentation

- 15 minutes presentation + 5 minutes Q&A
 - all members must participate
 - be prepared to defend assumptions and evidence
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9) Grading

- **Written report:** 80%
- **Presentation:** 20%
- Individual grades may be adjusted based on Evidence Packs and Q&A.

10) Bonus: Spec-Driven Development (+5% max)

To promote **research excellence in the AI era**, teams may earn up to a **+5% bonus** on the project grade for **well-executed specification-driven development**.

How the bonus is applied

```
project_score_with_bonus = min(project_raw_score × 1.05, 16)
```

The project score will **not exceed the full marks of the written report (16)** after applying the bonus.

11) Late Policy

- **≤ 1 day late:** max **80%**
- **≤ 3 days late:** max **50%**
- **> 7 days late:** **0**

Penalties apply **before** bonus calculation.

12) Resources

- Templates and examples are available on Canvas
 - Use primary sources and cite properly
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Instructor note: A small, well-tested system beats a large, unverified one.